

White paper: choosing a carryover-mitigation analysis strategy

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A two-page summary of ‘Choosing a carryover-mitigation analysis strategy for biomarker-treatment interactions in aggregated N-of-1 trials: a simulation-based tutorial and factorial evaluation’ (pmsimstats team, 2026-07-04).

Goal of the paper

In an aggregated N-of-1 trial the drug does not switch off when it is discontinued; its effect decays over days to weeks, so nominally off-drug occasions carry residual exposure. Treating them as cleanly off-drug mis-specifies the design matrix and attenuates the biomarker-treatment interaction test, which is the estimand these trials exist to serve.

The analyst therefore has to decide how to represent residual exposure in the linear predictor. The paper’s goal is to settle that decision empirically rather than by convention. Specifically, it aims to:

1. Compare three analyst-side representations of drug exposure under a fixed mixed model, and identify which is most robust to carryover.
2. Test whether the ranking is invariant to the assumed biomarker biology, the assumed decay shape, and the design, since a ranking that transfers would justify a single recommendation.
3. Quantify how much a wrong assumption actually costs, relative to other threats to power.

The motivation is a gap left by the companion paper, which showed that the carryover penalty depends on how the biomarker effect is encoded in the data-generating process (1-3% of power under Architecture A, direct mean moderation; roughly 31% under Architecture B, MVN differential correlation). That work fixed the decay shape at exponential and used a single analysis specification, leaving the joint sensitivity to mis-specification at *both* the data-generating and the analyst layer uncharacterized.

The three analysis specifications

All three share one model, `nlme::lme(Sx ~ bm * X + t)` with a random intercept and `corCAR1` residual correlation, and differ only in how the exposure regressor X is built:

- **A1 (binary).** $X = D_{it}$, on-drug indicator; carryover is ignored. The dominated reference.
- **A2 (exposure-weighted).** $X = Dbc_{it}$, a continuous predictor decaying off-drug under an assumed form and half-life. This is the current pmsimstats-ng default.
- **A3 (lagged-augmented).** $X = D_{it}$ plus a lagged just-off-drug nuisance term, the classical Jones-Kenward crossover device. No decay form is assumed.

Simulation setup

An ADEMP-structured factorial crossing three DGP decay forms (linear, exponential, Weibull at $k \in \{0.7, 1.0, 1.5\}$) with the three specifications, under both DGP architectures, three designs (CO, Hybrid, OL+BDC), two sample sizes ($N = 35, 70$) and three interaction strengths: 540 cells at 500 replicates each, plus five single-axis sensitivity blocks and a cluster-robust recalibration block. Common random numbers within each cell make the A2-versus-A1 contrast a paired comparison. Monte Carlo standard error on power is at most 0.022 and is reported with every estimate.

Results

A2 does not dominate. This is the paper’s central and deliberately negative finding. The exposure-weighted specification attains the highest power in only 68 of the 360 non-null cells (19%).

Where it does win, it wins clearly. Under Architecture B in the two designs with a blinded-discontinuation phase (Hybrid, OL+BDC), A2 leads A1 and A3 by about 10 percentage points at the reference cell (0.860 against 0.766 and 0.770 at $t_{1/2} = 1.0$), roughly five MCSEs. A higher-precision 24-cell rerun at 600 trials confirms this with a paired McNemar test: 13.7 points at the strongest OL+BDC cell, $p = 6.6 \times 10^{-17}$.

Where it loses, it loses badly. Under the classical crossover design A2 is markedly inferior (0.488 against 0.830 for A1). Under Architecture A it is marginally dominated throughout (Hybrid, $t_{1/2} = 1.0$: 0.960 against 0.972).

Mechanism. A2 recovers signal that carryover has moved into the attenuated off-drug correlation. Under Architecture A the treatment effect stays proportional at every occasion, so there is no attenuated correlation to recover and the simpler binary indicator is at least as good. Under the crossover design the within-subject correlation already carries most of that signal, so A2 has nothing to gain and its down-weighting of off-drug points only costs.

A3 is essentially A1. Its lagged term enters only as a main-effect nuisance covariate, so the tested interaction coefficient is identical to A1's; the two differ only through absorbed variance and track closely throughout. The Jones-Kenward expectation that a shape-free lagged term should beat a parametric predictor under misspecification is not supported in the range tested.

Where A2 leads, the lead is robust. Wrong assumed decay shape never pushes A2 below A1 or A3 (worst case: a linear truth). A two- to four-fold error in the assumed half-life moves A2 only from about 0.83 to 0.77. Stronger autocorrelation lifts all three specifications equally, leaving the 5-6 point lead ρ -independent; the joint $\rho \times t_{1/2}$ check finds the effects additive, not interactive.

Dropout dominates the specification choice. At 30% MCAR dropout all three methods collapse to 0.12-0.25 and become indistinguishable; the A2-minus-A1 gap shrinks from roughly 0.10 to 0.03. Severity-related (MAR) dropout is slightly worse than MCAR because the highest-signal patients leave first.

The test is mildly conservative, and fixable. Mean null rejection is 0.039 across the 540 null cells, an artefact of the corCAR1 working covariance that over-estimates the standard error by 6-10%. A CR2 cluster-robust standard error restores κ to about one and size to nominal (0.045-0.049), recovers two to five points of power, and leaves the ranking intact, at the cost of far fewer degrees of freedom (about 480 down to 23). This was validated only in the Architecture B / Hybrid / $N = 70$ cells.

What follows for practice

The widely applied default 'always use the exposure-weighted predictor' is only conditionally correct. Use A2 when the design has a discontinuation phase *and* the biomarker is believed to act through a correlation channel (Architecture B), and report it with a CR2 cluster-robust standard error. Under a crossover design, or when the biomarker mechanically scales the mean effect (Architecture A), the simple binary indicator is at least as good and considerably simpler to defend. A roughly guessed pharmacokinetic half-life is adequate wherever A2 is used.

The blunt practical message is that none of this competes with retention: anticipated dropout governs power far more than the carryover specification does, and should dominate design effort accordingly.

Scope and limitations

The numbers come from a single parameter set calibrated to a prazosin/PTSD trajectory, so absolute power values are illustrative and only the ranking is portable, and even the ranking is conditional on architecture and design. The sensitivity blocks vary one factor at a time and can miss factor interactions; the single joint check (S5) showed none. The CR2 recalibration is validated in one stratum only.

One caveat internal to the headline table deserves emphasis: bias, MSE, and coverage there are scored against a single calibrated target ($\theta_{\text{true}} = -3.6$), which is A2's estimand, while A1 and A3 estimate a different coefficient. Those columns are therefore not comparable across specifications; only power and Type I error support like-for-like comparison, and the paper says so explicitly.

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